

Differential assessment of effects of increasing model complexity in THM coupled models of the FE experiment at Mt. Terri

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ABSTRACT

The Full-scale Emplacement (FE) experiment is studied numerically using large 3D meshes with over one million elements and degrees of freedom. Non-isothermal Richards flow coupled with mechanics is the applied physical process model. Systematic model variations are performed to generate results for a systematic comparison to a large set of measurement data utilising statistical methods. These variations are based on several hypotheses concerning potential sources of uncertainty or mismatch between modelled and measured data. Among them are geometrical aspects as well as aspects of model complexity, such as including additional materials or physical processes. The geometrical aspects include the effects of varying the inner tunnel radius, changing the heater representation, examining the uncertainty of the sensor locations and investigating geometrical aspects of adding a shotcrete layer. Complexity is increased by including shotcrete and a simplified excavation damaged zone (EDZ) as additional materials, by accounting for a ventilation phase and by providing an excavation-induced non-homogeneous initial liquid pressure field. We discuss the nature and magnitude of the effects of each of these aspects as well as their combinations. Notably, we find a large influence of the geometrical aspects on the temperature results in the tunnel on the one hand and on the other the importance of the initial liquid pressure field in the Opalinus clay. Overall, the comparison to the measurement data can be improved by incorporating the investigated aspects into models of the FE experiment.

1. Introduction

In the Full-scale Emplacement (FE) experiment¹ at the Mont Terri Underground Rock Laboratory, an emplacement tunnel for high-level radioactive waste is investigated at full scale using the reference repository design of the National Cooperative for the Disposal of Radioactive Waste of Switzerland.² Within a 50 m long tunnel (2.5 to 2.7 m in diameter, partly supported by shotcrete), three cylindrical heaters mimicking waste canisters (4.6 m long and 1.05 m in diameter) have been placed centred lying on bentonite block pedestals. The rest of the tunnel volume has been backfilled with compacted granular bentonite up to the front end, that has been sealed with a concrete plug. During and after the tunnel has been excavated, ventilated, filled and sealed as well as during the phase where the heaters heat the structure, the development of temperature, liquid pressure and other quantities is studied extensively both in the bentonite and in the host rock Opalinus clay.^{3–5}

Task C of the Decovalex-2023 phase of the Decovalex project⁶ was initiated with the aim of studying the FE experiment numerically, with a focus on the increase in the liquid pressure in the Opalinus clay during the heating phase. In the course of analyses of large heater experiments,^{7–11} including the FE experiment,¹² certain features of the development of the liquid pressure in Opalinus clay have been shown to be impossible to capture by simple thermohydraulic (TH) or thermo-hydromechanical (THM) models. For example, commonly, the magnitude or timing of the peak and the rate of pressure dissipation cannot be satisfactorily reproduced over a range of observation points with a unique parameter set. Different hypotheses were proposed as potential explanations. In the work at hand, we investigate some of them systematically in a stepwise manner. Although in reality, this has been an iterative process of hypothesis formation, model development, calibration and comparison, we present the final set of relevant steps in this course in a linear way with increasing complexity.

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Different categories of uncertainty have to be addressed for complex THM modelling tasks, such as the degree of detail to which materials and processes have to be included, material parameter values, geometrical aspects, discretisation requirements, numerical settings, model parameters, etc. This work focuses on the effects of selected geometrical aspects, the choice of representing specific materials and processes, and to a lesser extent on parameter identification. Readers interested in the latter aspect may refer to Ref. 12.

We carried out extensive finite element analyses with large 3D meshes with more than one million elements and degrees of freedom. However, large models are by far not the biggest challenge in simulating in situ experiments such as this one. The FE experiment is equipped with hundreds of sensors that collect measurements over years. Incorporating such data into computational models represents another significant challenge. To provide a more comprehensive evaluation of the model variants studied here, extending beyond a comparison of just a few selected liquid pressure curves, data from a large set of sensors was used and integrated into appropriately defined error metrics. The established analysis workflow, while far from ideal, allows for systematic comparisons of different variants in an objective, automated, reproducible, but still flexible way.

Among the specific geometrical settings, we will discuss the effects of varying the inner tunnel radius, changing the heater representation, the uncertainty of the measurement point coordinates, and geometrical aspects of adding a shotcrete layer. When it comes to adding specific materials and processes, in particular, we look at including shotcrete and a simplified excavation damaged zone (EDZ) as additional materials, as well as a ventilation phase and a non-homogeneous initial liquid pressure field as extensions of the represented complexity.

For all numerical analyses, the non-isothermal hydromechanical theory with Richards' assumption (TRM¹³) was used in its implementation in OpenGeoSys 6 (OGS-6). Since the FE experiment does not include extremely high temperatures or severely elevated gas pressures, the TRM model can be applied to describe the coupled thermo-hydromechanical processes at lower computational costs than a full multiphase flow process model.^{14,15}

After having performed 2D benchmarks, 3D blind predictions and parameter variation studies, as outlined in Fig. 1, a base parameter set was selected to examine the effects of slightly varying geometry and increased complexity. For brevity, the 2D benchmarks and 3D blind predictions will be skipped here as they have been presented elsewhere.^{16,17} The parameter variation studies will be covered briefly to the extent required to justify the selection of the parameter set, which was the basis for the geometry and increased complexity studies.

The paper thus investigates the effects of several aspects of uncertainty based on the aforementioned hypotheses that have been grouped into geometry studies on the one hand and increased complexity studies on the other. The geometry studies examine aspects of the inner radius of the tunnel, the representation of the heater, the uncertainty of the coordinates of the measurement points, and the geometrical representation of the shotcrete. The increased complexity studies, on the other hand, include the presence (or absence) of shotcrete, the presence and representation of a simplified EDZ, the influence of a ventilation phase, and of a non-homogeneous initial liquid pressure field.

With the aim in mind to reproduce the measurement data as good as possible, the above-mentioned uncertainties are hurdles that have to be taken with limited opportunities of reducing the uncertainties themselves. With the work at hand, we provide the knowledge gained about their relative importance, as well as the way and magnitude in which they might affect the results.

2. Methods

We structure the methods into brief sections that present the theory used (Section 2.1), general modelling choices (Section 2.2), information about the measurement data (Section 2.3) as well as an overview of

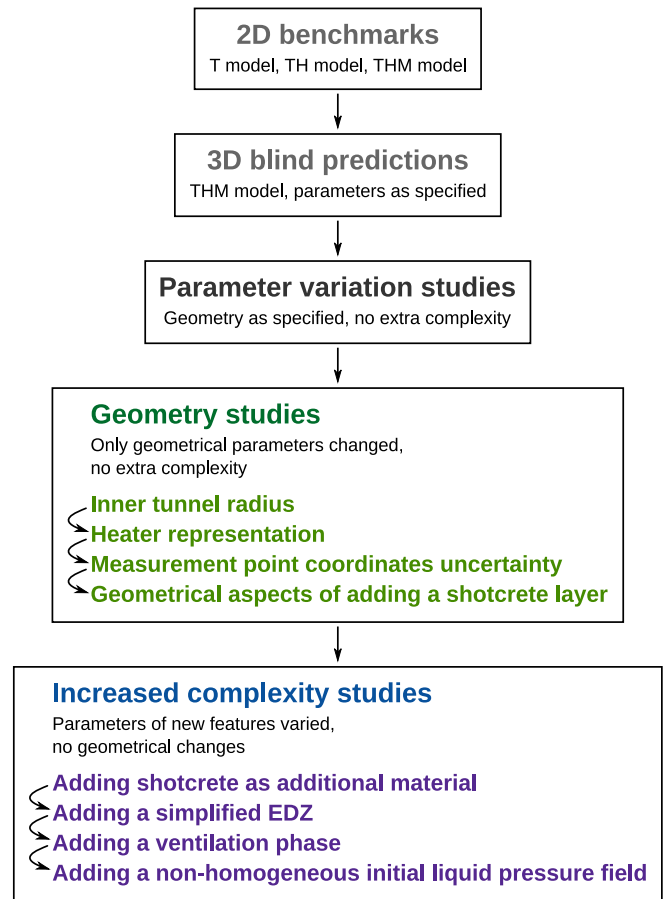


Fig. 1. Workflow for the modelling studies as presented here. While in reality, the process was iterative with various loops, parallel developments and dead ends, the relevant steps are shown here in a linear way with increasing complexity.

the parameter variation studies (Section 2.4). These are followed by two more detailed sections on the geometry studies (Section 2.5) and the studies on increased complexity (Section 2.6). The latter two have their own detailed structure, which is provided at the beginning of Section 2.5.

2.1. Theory

The theory of the TRM process model used is based on the non-isothermal Richards flow with mechanics. Richards' assumption implies that the mobility of gases in unsaturated porous media is always large enough to allow sufficiently fast drainage. Under such circumstances, the gas pressure does not increase significantly over drained levels (usually considered atmospheric), so that the capillary pressure equals the negative liquid pressure.

The process model is based on the balance equations for energy, mass and linear momentum, which will be presented here only briefly for completeness. For a detailed description, refer to Ref. 15.

Energy balance

The energy balance of the porous media is formulated as

$$Q_T = (\rho c_p)_{\text{eff}}' (T)'_S - \text{div}(\lambda_{\text{eff}} \text{grad } T) + (c_{pL} \rho_{LR} \tilde{w}_{LS} + c_{p,\text{vap}} \rho_{\text{vap}} \tilde{w}_{\text{vap},S}) \cdot \text{grad } T, \quad (1)$$

where Q_T is the source term, $(\rho c_p)_{\text{eff}}$ is the volumetric heat capacity, T is the temperature, λ_{eff} is the effective heat conductivity tensor, c_{pL} is the specific heat capacity of water, ρ_{LR} is the liquid density, $\tilde{w}_{LS}$ is the Darcy flux of the liquid, $c_{p,\text{vap}}$ is the specific heat capacity of water

vapour, ρ_{vap} is the vapour mixed gas density, and $\tilde{\mathbf{w}}_{\text{vap},S}$ is the Darcy flux of the water vapour,

$$\tilde{\mathbf{w}}_{\text{vap},S} = -D_{v,\text{eff}} \text{grad } \rho_{\text{vap}}^W = -D_{v,\text{eff}} \left(\frac{\partial \rho_{\text{vap}}^W}{\partial p_{\text{LR}}} \text{grad } p_{\text{LR}} + \frac{\partial \rho_{\text{vap}}^W}{\partial T} \text{grad } T \right)$$

and $D_{v,\text{eff}} = \phi(1 - S_L)D_v$,

where $D_{v,\text{eff}}$ is the effective water vapour diffusion coefficient, ρ_{vap}^W is the water vapour density, p_{LR} is the liquid pressure, and D_v is the relative vapour diffusion coefficient.

Mass balance

The mass balance of the pore water reads

$$\begin{aligned} Q_H = & \left(\rho_{\text{LR}} - \rho_{\text{vap}}^W \right) \left[\phi - p_{\text{cap}} S_L (\alpha_B - \phi) \beta_{p,\text{SR}} \right] (S_L)'_S \\ & + \rho_{\text{LR}} S_L \left[\phi \beta_{p,\text{LR}} + S_L (\alpha_B - \phi) \beta_{p,\text{SR}} \right] (p_{\text{LR}})'_S \\ & + \phi \left[(1 - S_L) \frac{\partial \rho_{\text{vap}}^W}{\partial p_{\text{LR}}} \right] (p_{\text{LR}})'_S + \text{div} \left(\rho_{\text{LR}} \tilde{\mathbf{w}}_{\text{LS}} + \rho_{\text{vap}}^W \tilde{\mathbf{w}}_{\text{vap},S} \right) \\ & + S_L \rho_{\text{LR}} \alpha_B \text{div}(\mathbf{u})'_S \\ & + \left[\phi(1 - S_L) \frac{\partial \rho_{\text{vap}}^W}{\partial T} - \rho_{\text{LR}} S_L \left[\phi \beta_{T,\text{LR}} + (\alpha_B - \phi) \alpha_{T,\text{SR}} : \mathbf{I} \right] \right] (T)'_S, \end{aligned} \quad (3)$$

where Q_H is the source / sink term, ϕ is the porosity, p_{cap} is the capillary pressure, S_L stands for the liquid saturation, α_B is the Biot's coefficient, $\beta_{p,\text{SR}}$ is the solid compressibility, $\beta_{p,\text{LR}}$ is the liquid compressibility, $\mathbf{u}$ is the displacement, $\beta_{T,\text{LR}}$ is the linear thermal expansivity of the liquid, $\alpha_{T,\text{SR}}$ is the linear thermal expansivity tensor of the solid, and $\mathbf{I}$ is the identity tensor.

The porosity variation is considered by the following relation expressed in primary variable rates:

$$(\phi)'_S = (\alpha_B - \phi) \left[\text{div}(\mathbf{u})'_S - \alpha_{T,\text{SR}} : \mathbf{I}(T)'_S + \beta_{p,\text{SR}} (\chi p_{\text{LR}})'_S \right], \quad (4)$$

with the Bishop's model formulated as saturation cut-off $\chi(S_L) = \begin{cases} 0 & \text{if } S_L < 1, \\ 1 & \text{else.} \end{cases}$

Linear momentum balance

The linear momentum balance of the porous media is written as

$$0 = \text{div}(\sigma_{\text{eff}} - \alpha_B \chi p_{\text{LR}} \mathbf{I}) + [(1 - \phi) \rho_{\text{SR}} + S_L \phi \rho_{\text{LR}}] \mathbf{b}, \quad (5)$$

with the effective stress tensor σ_{eff} , the solid density ρ_{SR} , and the body force vector $\mathbf{b}$.

The aim of this contribution is to discuss modelling choices and other uncertainties in a common context. To achieve this, some details of the model setup, such as the initialisation phase of the liquid pressure, have to be omitted here because the presentation would otherwise become too lengthy. For details, we refer the reader also to Refs. 12, 16.

2.2. Computational approach

The open-source software OpenGeoSys^{18,19} for thermo-hydro-mechanical-chemical problems is based on the finite element method (FEM). We use the OGS-6.4 version and therein the ThermoRichardsMechanics process, which utilises a monolithic scheme for coupled thermo-hydro-mechanical processes. The balance equations implemented in this process are those presented in Section 2.1. The time discretisation is a backward Euler scheme and the time step length is adaptively chosen based on the number of nonlinear iterations required for convergence. The initial time step is set to 5 d, the minimum time step to 1 d and the maximum time step to 20 d. The used solver is a Newton–Raphson nonlinear solver, the error of which is controlled by the L^2 -norm of the component-wise solution increment as convergence criterion.

Most of the parameter variation studies have been performed using a 200 m cubic domain and an unstructured mesh with 1.27 M

tetrahedral elements and 1.05 M degrees of freedom. The formation of Opalinus clay in Mont Terri is less than 200 m. Using nevertheless this domain size is intended to reduce boundary stiffness effects. Although most of the geometry and complexity studies also used the 200 m cubic domain and unstructured meshes, the number of tetrahedral elements increased to 2.91 M, which means 2.38 M degrees of freedom. Fig. 2 gives some impressions of the structure and the degree of refinement of the second group of meshes. Enclosed by the host rock Opalinus clay, the technical structure is shown with its different materials, namely EDZ, shotcrete, granular bentonite, concrete for the plug and the pedestal bases, bentonite blocks for the pedestals, and carbon steel for the heaters.

Using meshes with millions of elements and degrees of freedom and adapting them to the manipulation of the underlying geometry are not straightforward in 3D. A minimum degree of resolution is necessary to adequately represent certain features of the structure that might have significant effects on the behaviour of the entire model. In consequence, using detailed, computationally large 3D meshes requires an efficient process model implementation, while geometrical variations require automated and suitably parameterised meshing scripts. Using these, in the early phase of Decovalex-2023, we did mesh convergence studies in 2D and later used also the findings of other participants in Task C. The meshes we used in 3D were designed as detailed as was still possible to handle without too many difficulties.

As already stated before, parameter variation studies were performed before examining the effects of slightly varying geometry and increased complexity. The starting point for these parameter variations was the task specification documented in the report by Ref. 16.

2.3. Measurement data

As mentioned earlier, treating measurement data in models of in situ experiments is a challenge in itself. The data provided from the FE experiment, with which we compared our model results, are measurements with a frequency of one per hour. They originate from 207 out of the over 1600 sensors and cover six variables: temperature (T), liquid (pore water) pressure (p_L), relative humidity (RH), volumetric water content (VWC), displacement ($\mathbf{u}$) and strain (ϵ). There have been 86 279 time stamps provided in total and about 10 M non-NA data points. If we restrict our calculations according to the task specification to the first five years after the onset of heating of the first heater, we are left with 43 831 time stamps and about 7 M non-NA data points. After a rough cleaning (see below), we still have 43 831 time stamps and about 6.5 M non-NA data points.

Together with the data, some documentation was provided, in total 1326 pages of text, including a paper and four reports. The data had to be extensively processed to be useful for integration in or comparison to numerical models. This included resolving issues about

- contradictory and different geometry compared to the specification;
- unclear (close to the tunnel wall) or missing (ϵ) sensor positions;
- missing or contradictory information about sensor ranges, their failures and resets;
- raw data including values, that are out of range, outliers, measurement errors, sensor failure or various missing value identifiers;
- double sensors with contradictory data;
- contradictory heater power schemes concerning the boundary conditions;
- and unclear directions of the directional variables $\mathbf{u}$ and ϵ .

To limit the complexity of this task and to avoid uncertainty concerning the mechanical part, in the presented work we did not concentrate on the data from the displacement and strain sensors. Data cleaning was done individually for variables, boreholes, or even single sensors and

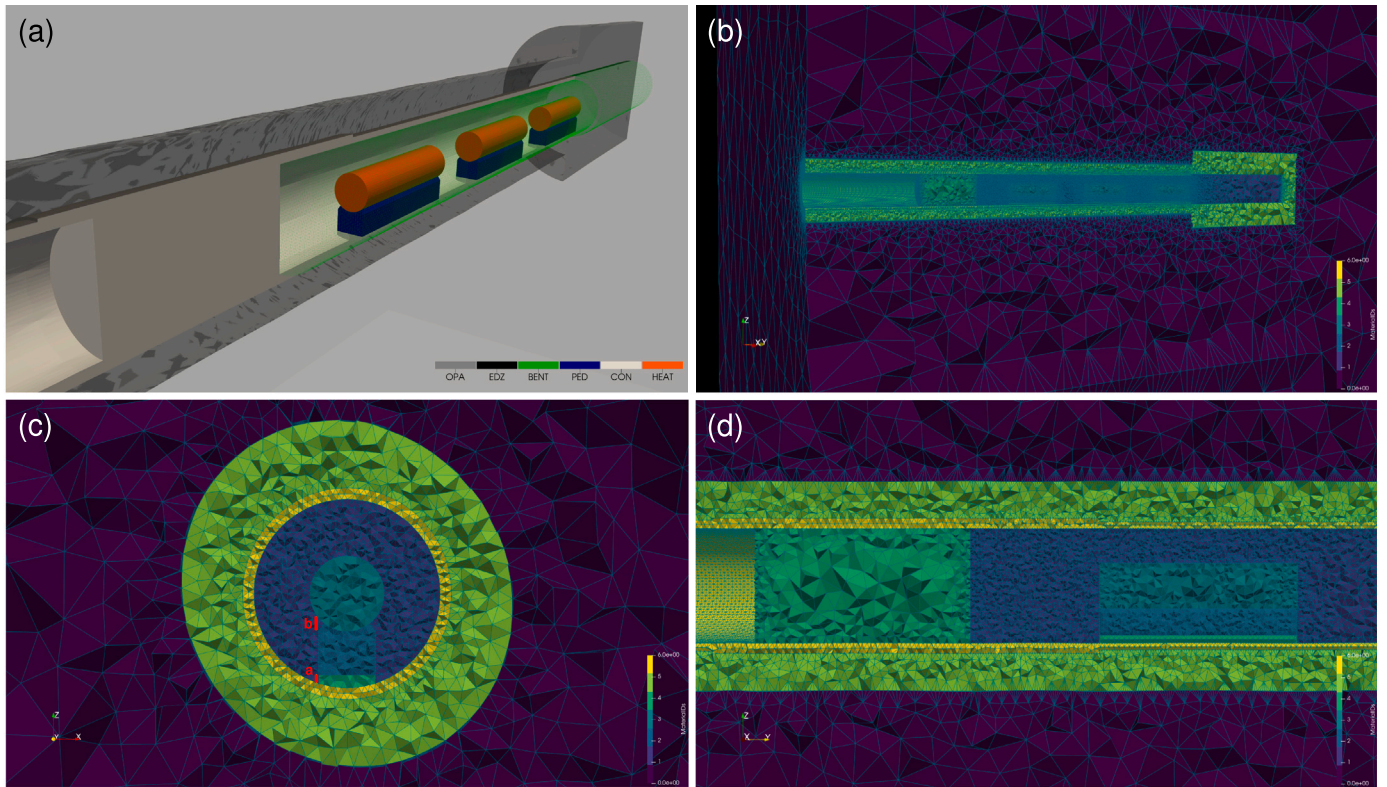


Fig. 2. Close-ups of (a) geometry and (b) – (d) mesh used for the geometry and increased complexity studies (2.91 M tetrahedral elements). Shown are especially (a) and (b) perspective views of the tunnel region to illustrate (a) the main structural elements and (b) the degree of refinement towards the tunnel interior, (c) cross-section (i.e. xz plane) through the middle heater and (d) longitudinal section (i.e. yz plane) through the tunnel centre to show the different materials. These are Opalinus clay in red–purple, EDZ in light green, shotcrete in yellow, granular bentonite in blue–purple, concrete for the plug and the pedestal bases in green, bentonite blocks for the pedestals in blue, carbon steel for the heaters in blue–green in the colours of (b) – (d). The shotcrete can be seen to vary in thickness along the tunnel length in (a) and (d). In (a) and (b), the increased size of the EDZ can be seen where there is no shotcrete. Additionally, in (c) two geometric lengths are marked in red, that are mentioned in Table 3 and shall illustrate what is meant by “Outer (i.e. Minimum) Height Pedestal Base” (a) and “Vert. Dist. to Heater of Edge Pedestal Vert. to Slope” (b).

by looking at figures of the data and judging what should definitively be ignored, while being careful to keep data points in case of doubt.

For a meaningful comparison between model and data, due to the differences between real and idealised modelled geometry, some sensor positions had to be adjusted in the model. Sensors that in reality were sitting close to the real tunnel wall were moved to a distance of 10 cm inside of the idealised tunnel wall, while keeping their angle and position along the tunnel axis unchanged. In this way, we ensured that they lie in the correct material domain despite geometrical simplification. Sensors that were close to a real pedestal were placed in the direction of the tunnel length at 12 cm distance from their idealised pedestal, keeping their angle and radius relative to the tunnel as they were. If both conditions held, both repositionings were done. To give an impression of the available measurement points, we exemplarily provide a categorisation for the temperature measurement points in Table 1 and sketch the locations of these categories in Fig. 3. At points falling into the categories OpPa and OpPe, also pressure measurements were available.

Table 1
Categories of temperature measurement points.

Name	Count	Location
HeBe	12	directly on the heaters in granular bentonite
HeBl	3	directly on the heaters in bentonite blocks
TuBe	12	20 cm away from the heaters in granular bentonite
TuBl	3	20 cm away from the heaters in bentonite blocks
WaBe	10	10 cm away from the tunnel wall in granular bentonite
OpPa	18	in Opalinus clay parallel to the bedding
OpPe	18	in Opalinus clay perpendicular to the bedding

Shown are the category names together with numbers and locations of the included temperature measurement points.

2.4. Parameter variation studies

The THM model of the FE experiment used for the parameter variation studies includes only the four materials Opalinus clay (OPA), granular bentonite (Bent), bentonite blocks (Block) and concrete as well as water. Still, in total 75 material parameters as well as complex initial and boundary conditions have to be provided. The model parameters and characteristics that have been varied for at least one material are briefly listed in Table 2.

Among the performed runs were some in which we changed just one single parameter at a time, as well as many others in which we varied several (physical) parameters simultaneously. These combinations furthermore split in parameter sets that act as proxies for anticipated missing features, and physical parameter sets. The proxies can give results that might fit well to the observations but are without physical justification themselves. In contrast, the physical parameter sets might not give the best fit to the observations for various reasons but can themselves be regarded as real possibilities of parameter settings.

The model results of these variation runs were compared at all observation points at which measurement data of at least one variable was available. There were 219 variation runs with 200 location/variable pairs and about 100 time steps of varying length each. All of them should be compared with each other and with the measurement data (43 831 time stamps). To handle this task appropriately, statistical methods were applied as described below.

First, all the model results at all sensor positions were interpolated from the model time steps to the measurement data time stamps. Then, comparisons were made only relative to the starting values. In the course of the modelled time frame, large relative changes occur in some

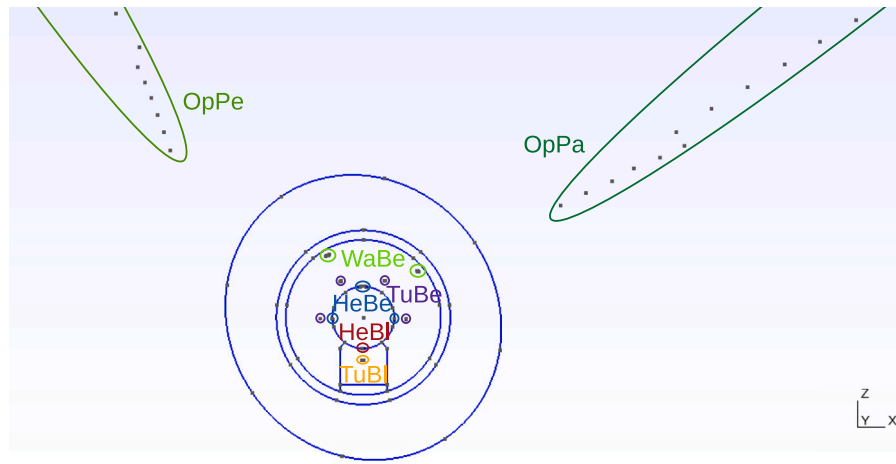


Fig. 3. Positions of the categories listed in Table 1 for the available temperature measurement points. The geometry from outside to inside of EDZ, shotcrete, tunnel, pedestal base, pedestal and heater is included for reference.

Table 2

Model parameters varied in the course of the parameter variation studies.

Numerical parameters	Physical parameters
Element count	Thermal conductivity
Domain size	Dry bulk density
Linear solver	Porosity
Error tolerances of linear solver	Intrinsic permeability
Error tolerances of non-linear solver	van Genuchten entry pressure
Initial time step size	van Genuchten parameter n
Minimum time step size	Young's modulus
Maximum iteration steps non-linear solver	Linear thermal expansivity
Maximum iteration steps linear solver	Biot coefficient
Number of CPUs	Initial stress
Number of parallel runs	Water density model
	Vapour diffusion model
	Temperature initial conditions
	Pressure initial conditions
	Temperature boundary conditions
	Pressure boundary conditions
	Latent heat
	Heat source representation
	Gas phase representation

Listed are those numerical and physical parameters of the TRM model that were varied in the course of the parameter variation studies.

parts but only small ones in others. Thus, it was beneficial to split the time series into shorter overlapping sections for separate treatment. To be precise, all time series were split into sections using running 0.5 a windows, with shorter 0.25 a sections at the start and end. Then, the Index of Agreement,²⁰

$$IA = \begin{cases} 1 - \frac{\sum_{i=1}^n |M_i - D_i|}{2 \cdot \sum_{i=1}^n |D_i - \bar{D}|}, & \text{if } \sum_{i=1}^n |M_i - D_i| \leq 2 \cdot \sum_{i=1}^n |D_i - \bar{D}| \\ \frac{2 \cdot \sum_{i=1}^n |D_i - \bar{D}|}{\sum_{i=1}^n |M_i - D_i|} - 1, & \text{if } \sum_{i=1}^n |M_i - D_i| > 2 \cdot \sum_{i=1}^n |D_i - \bar{D}| \end{cases} \quad (6)$$

where M stands for model and D for data, was calculated for every sensor (i.e. location/variable pair), every time series section, every model run, and every dimension in space (for displacement and strain). Calculating the IA , there are comparisons of individual data values D_i to the mean of the data time series $\bar{D}$ used, the so-called comparison to the perfect model. Consequently, large variations of the differences to the mean will cause the IA to decrease in information content. This is the reason for splitting the 5 a time series into pieces, as smaller pieces reduce the effect of non-linearities.

There are different statistical measures that can be used, and all have their shortcomings and advantages. The choice of using the IA

was promoted by its non-dimensionality, i.e. that one can automatically compare variables of different units, the possibility to feed it with negative as well as positive values, its symmetry for over- and underprediction, its boundedness, and its straightforward way of interpretation. If interested in this in more detail, the reader is referred to Ref. 20, but also e.g. Refs. 21–23.

We were not interested in the performance of a configuration just at one single point for just one single variable during a short time period. But instead, we were looking for one that would ideally give good results for many sensors over most of the modelled time frame. The next step was therefore the aggregation of the single IA values. In an exploratory way, we used several possible options, to be able to decide about the most suitable method later on. In particular, the resulting 4D array (sensor, time series section, model run, space dimension) of single IA values was aggregated by calculating all quartiles and the mean over various different sensor groups and several different time slices for all model runs and all dimensions (5D: sensor group, time slice, model run, space dimension, aggregation method).

The sensor groups included the partition of the sensors according to their measured variable, their surrounding material, their position compared to the direction of the bedding, their radius, their position to each of the three coordinate axes, all sensors together, and all but excluding the mechanical variables, as well as all possible combinations thereof. The time slices, on the other hand, combined the time series sections all together, only all the short subannual sections, only the 5 a section, or all the short sections, which include time stamps of year 1 or 2 or ... or 5, respectively.

For every combination of the above-mentioned choice of aggregation method, sensor group and time slice, an aggregated value of the IA was created. The variety of choices still allows for various comparison possibilities between different model runs. After checking and comparing the aggregated IA values, the final choice of method was determined. We decided to use all sensors excluding the mechanical ones and all subannual time series sections together and to use means with equal weights over both the sensors and the time series sections. This configuration turned out to be the most stable measure of model performance, being the most suitable one in light of the uncertain quality of the measurement data. At the same time, it enabled many sensors and time series sections to be judged at once, not weighing them in the light of missing information about the goodness of the individual data points. Finally, since there was only one value per configuration left that way, the “best” model fit would be the one with the highest aggregated IA value among all different runs.

Using this method, a single “best” model was picked for further use. Additionally, the properties of models that had good fits according

to this very choice as well as similar others were compared, too, to identify patterns that lead to a better fit. In general, sensitivities to single parameters were studied at this stage, but also relevant features not included in the current setup were revealed due to the use of the proxies.

2.5. Geometry studies

Deviations between measurement data and model results remained despite extensive parameter variations. This raised questions about the influence of specific model settings on the results. But also the influence of materials that were originally excluded from the model gained interest. Questions primarily related to geometrical aspects are grouped in the current section under the name of geometry studies, while others will be grouped under the name of increased complexity studies in Section 2.6.

The ideas followed and presented here under the name geometry studies are the effects of varying the inner tunnel radius (Section 2.5.1), changing the heater representation (Section 2.5.2), examining the measurement point coordinates uncertainty (Section 2.5.3), and investigating geometrical aspects of adding a shotcrete layer (Section 2.5.4). The aspects of the presence of shotcrete (Section 2.6.1) and of the presence and representation of a simplified EDZ (Section 2.6.2) as additional materials, the influence of a ventilation phase (Section 2.6.3) and of a non-homogeneous initial liquid pressure field (Section 2.6.4) will be presented as part of the increased complexity studies in the next section.

The geometry of a 3D model of the FE experiment is anticipated to need to incorporate two diametral features for a reasonably good accuracy of specific modelling results: a large volume of hundreds of meters in all directions as well as fine details of just a few centimeters in size. It will thus need detailed meshing in some regions. But it cannot have the same degree of detail in the whole volume, simply because of the resulting number of elements and the computational effort to handle them. It is also not wise or even impossible to try to extrude a 2D geometry that allows for all the needed 3D features. Instead, the geometry has to be adjusted in a way that fulfils well the structural needs but does not collide with the computational possibilities at the same time. Most of the aforementioned aspects in question relate to the geometry of the model and require several variations. Geometry and meshing used for these studies were thus scripted to be almost completely parametric and that way allowed to perform the desired manipulations. We used Gmsh²⁴ version 4.11.1, 2022, for this task.

In addition to that, the features of this second group of geometry/meshes include volumetric carbon steel heaters, an inner tunnel radius that fits the range provided by the documentation of the FE experiment, an improved pedestal geometry, concrete pedestal bases, shotcrete of varying thickness, a simplified anisotropic EDZ, and realistic measurement point locations. As mentioned earlier, Fig. 2 provides some close-ups of one geometry and mesh of this group.

With this, systematic testing of the influence of varying the inner tunnel radius, changing the heater representation, the measurement point coordinates uncertainty, the shotcrete presence and representation, the EDZ presence and representation, a ventilation phase, and a non-homogeneous initial liquid pressure field were all possible. Thus, the effects of a slightly varying geometry and increased complexity in terms of additionally included materials, which were originally excluded from the model, could now be studied.

The parameter set chosen from the parameter variation studies and used together with the second group of meshes as basis for the following studies is briefly presented here. Table 3 provides those parameter values that are different from those given in the specification by Ref. 16. The value used for the intrinsic permeability in Opalinus clay parallel to the bedding is 0.125 times the value given there, while perpendicular to the bedding, the factor is 0.375.

Table 3

Model parameters that differ from the specification.

Parameter	Changed value
Domain size	200 ³ m ³
Inner Radius Tunnel	1.315 m
Outer (i.e. Minimum) Height Pedestal Base	11.5 cm
Vert. Dist. to Heater of Edge Pedestal Vert. to Slope	18.5 cm
Heat source	volumetric
Temperature IC OPA	16.5 °C
Temperature IC else	18.0 °C
Thermal conductivity Dry Block	0.8 W m ⁻¹ K ⁻¹
Thermal conductivity Saturated Bent	1.0 W m ⁻¹ K ⁻¹
Thermal conductivity Curve Vapour	(24.57, 26.66, 28.83, 31.09, 33.43) mW m ⁻¹ K ⁻¹ at (100, 125, 150, 175, 200) °C
Linear thermal expansivity OPA	10 ⁻⁵ K ⁻¹
Specific heat capacity Vapour	1850 J kg ⁻¹ K ⁻¹
Relative humidity IC Bent (rounded to 1 digit)	30.3 %
Relative humidity IC Block (rounded to 1 digit)	69.5 %
Pressure BC pos./neg. z-axis	Dirichlet OPA 0 m
Entry pressure Bent	10 MPa
van Genuchten n Bent	1.67
Intrinsic permeability OPA	6.25 · 10 ⁻²¹ m ²
Intrinsic permeability OPA ⊥	3.75 · 10 ⁻²¹ m ²
Porosity OPA	0.17
Porosity Bent	0.45
Biot coefficient OPA	0.6
Young's modulus OPA	4.8 GPa
Young's modulus OPA ⊥	2.4 GPa
Stress x OPA	4.0 MPa
Stress y OPA	2.2 MPa

Listed are those parameters of the TRM model that are different from the ones given in the specification. The symbols || and ⊥ refer to the orientation with respect to the bedding plane.

The used water density model (rounded to three digits) is

$$\rho_{LR} = 1.019 \cdot (5.388 \cdot 10^{-6} \text{ K}^{-3} \cdot T^3 - 8.619 \cdot 10^{-3} \text{ K}^{-2} \cdot T^2 + 3.446 \text{ K}^{-1} \cdot T + 5.926 \cdot 10^2) \cdot \exp(4.65 \cdot 10^{-10} \text{ Pa}^{-1} \cdot (\max(0 \text{ Pa}, p_{LR}) - 10^5 \text{ Pa})) \text{ kg m}^{-3}, \quad (7)$$

while the used vapour diffusion model is

$$D_v = 2.16 \cdot 10^{-5} \text{ m}^2 \text{ s}^{-1} \cdot (T/273.15 \text{ K})^{1.8} \cdot \tau \cdot \phi \cdot (1 - S_L) \text{ with } \tau = 0.8 \text{ (FEBEX)}, \quad (8)$$

which was presented in Ref. 25 as Eq. (4.14).

2.5.1. Inner tunnel radius

The specification of Task C made use of an idealised geometry of the FE experiment. One aspect of this is an inner tunnel radius that is slightly smaller than the range provided for the real tunnel geometry by the documentation of the FE experiment. As the tunnel is filled with unsaturated bentonite, while the surrounding host rock is saturated Opalinus clay, the influence of this geometrical choice must be quantified.

To test the influence of the inner tunnel radius on the modelling results, two mesh variants were created. One used the value of 1.24 m for the inner radius of the tunnel provided by Ref. 16 and one used a value of 1.315 m. The latter value was also chosen as the standard for the parametric meshes used and discussed in the following. It is in line with the documentation of the FE experiment, which gives a range from 1.25 to 1.35 m for the radius of the real tunnel geometry.¹

2.5.2. Heater representation

A heater can be represented in various ways, such as via a specified heat flux at the heater surface, as a volumetric heat source with its own set of thermal properties or even with its internal structure fully resolved.

We tested the influence of using meshed carbon steel heaters with volumetric heat source terms, and a variant with holes in the mesh in place of the heaters combined with surface heat fluxes, as proposed by Ref. 16. We did not test heaters with internal structure but applied

Table 4

Model parameters (composite) used for the uniform carbon steel heaters.

Parameter	Used value
Thermal conductivity	$33 \text{ W m}^{-1} \text{ K}^{-1}$
Linear thermal expansivity	$1.4 \cdot 10^{-5} \text{ K}^{-1}$
Specific heat capacity	$502.416 \text{ J kg}^{-1} \text{ K}^{-1}$
Grain density (rounded to 3 digits)	$1276.629 \text{ kg m}^{-3}$
Young's modulus	210 GPa
Poisson ratio	0.295

Listed are those parameters of the TRM model that were used for the uniform carbon steel heaters as composite material.

effective parameters to uniform heater volumes. The parameters of these uniform heaters are listed in Table 4.

2.5.3. Measurement point coordinates uncertainty

As written earlier, the difference between real geometry and idealised tunnel is especially an issue, when it comes to the locations of measurement points close to material boundaries. For this reason, in the study on the measurement point coordinates uncertainty, two complete sets of measurement point coordinates from the FE experiment were used and compared.

The specification by Ref. 16 provided the first set of measurement point coordinates. Its locations had been adjusted to the idealised model geometry of the tunnel, and were additionally idealised themselves using standardised distances. In detail, were they sitting directly at the heaters, 20 cm away from them, and 10 cm away from the tunnel wall. These measurement point locations differ maximally by about 15 cm from the measurement point locations of the FE experiment in reality, which is about 12 % of the inner tunnel radius itself. Therefore, the question was raised as to whether these differences could cause differences in the performance of the model results compared to the measurement data. This was assumed because the model results were evaluated at different locations compared to where the measurement data was obtained.

With the parametric mesh, we tested the influence of this choice by additionally using a second set of measurement point coordinates. This set resembled, as far as possible, the coordinates of the measurement point locations of the FE experiment. It was only adjusted to the idealised model geometry when it became inevitable, using distances of 2 cm from the heater or the tunnel wall, respectively.

Artificial measurement point coordinates

In a second systematic investigation independent of the actual sensor locations and geometrical simplifications, we created a set of artificial measurement points in the plane through the centre vertical cross-section of the second heater. Their positions are at radii 0.55 m (close to the heater; HeTo and HeRi), 0.8 m (in the tunnel; TuTo and TuRi), 1.2 m (close to the tunnel wall; WaTo and WaRi), all along the vertical and horizontal directions, and at 4 m and 10 m (in the OPA; OpPa and OpPe), both parallel and perpendicular to the bedding. Fig. 4 shows the locations and names of these points. Then, the positions of the artificial measurement points were varied by ± 1 , ± 2 , ± 5 and ± 10 cm in each of the three spatial directions, so that differences in the model results due to these spatial variations could be revealed.

2.5.4. Geometrical aspects of adding a shotcrete layer

In principal, we are interested in the question of what effect adding a shotcrete layer at the boundary between the tunnel and the Opalinus clay would have on the model results. But there is a geometrical issue, whose effect should be investigated and quantified separately from the effect of adding the material itself: When a model with shotcrete is compared to one without it, the volume that is occupied by shotcrete in the former will be occupied by another material in the latter. And with the same argument as for the inner tunnel radius (Section 2.5.1), the influence of the choice, which material this is, has to be quantified.

If one decides to use a part of the Opalinus clay to add the shotcrete layer,¹⁶ this means in essence adding it outside of the tunnel and will thus increase the engineered part of the model domain. However, judging from the real coordinates of the measurement points but also taking physical consequences into account, it might be more appropriate to keep the engineered part of the barrier the same size. This will mean adding it inside of the tunnel and thus using a part of the bentonite for it. The difference between having bentonite or Opalinus clay in the volume, where shotcrete could be, can be addressed by increasing the tunnel radius itself. Thus, one can also address the addition of shotcrete outside the tunnel as a combination of increasing the tunnel radius and afterwards adding shotcrete inside the tunnel.

Consequently, the volume of shotcrete being Opalinus clay, the volume of shotcrete being bentonite, and it being shotcrete are three situations that were compared against each other by manipulating the parametric mesh in the necessary ways to resemble the mentioned geometries. For simplicity, at this stage, we used the same parameters for the shotcrete as were provided for the concrete by Ref. 16.

2.6. Increased complexity studies

Aside from uncertainties related to parameter values and model geometries, the selection of features to represent in a model can largely determine the results, but also add significantly to complexity. After having examined the effects originating from geometry presented in the previous section, the model complexity could now be increased stepwise, building upon the base model. To be precise, considering shotcrete (Section 2.6.1) and a simplified EDZ (Section 2.6.2) as materials that were originally excluded from the model, as well as including an additional ventilation phase prior to the heating phase (Section 2.6.3) and a non-homogeneous initial liquid pressure field for Opalinus clay (Section 2.6.4) were among the investigated features.

2.6.1. Adding shotcrete as additional material

To bring the liquid pressure results closer to the measurements, it was assumed that adding a shotcrete layer to the model might have a positive effect by acting as a barrier between bentonite and Opalinus clay. The shotcrete supports the tunnel in the FE experiment, but was not included in earlier models within this modelling task.

Consequently, we compared a model with shotcrete to one without it. While the inner tunnel radius was kept constant as described in Section 2.5.1, the thickness of the shotcrete layer varies over the tunnel length. Consequently, also the overall diameter of the engineered structure varies with it (see also Fig. 2). The parameter values chosen for shotcrete that are different from those used for concrete (provided by Ref. 16) are shown in Table 5.

2.6.2. Adding a simplified EDZ

Another hypothesis was that an EDZ could improve the comparison of the liquid pressure development to the measurement data. During

Table 5

Model parameters used for shotcrete that differ from those used for concrete.

Parameter	Used value
Thickness along tunnel sections	(19, 24, 16) cm at (4.7–13.7, 13.7–22.7, 22.7–42.7) m
Thermal conductivity Dry	$1.06 \text{ W m}^{-1} \text{ K}^{-1}$
Thermal conductivity Saturated	$1.70 \text{ W m}^{-1} \text{ K}^{-1}$
Grain density	2700 kg m^{-3}
Dry bulk density	2076.3 kg m^{-3}
Volumetric water content	4.7 vol%
Saturation (rounded to 3 digits)	0.203
Pressure IC (rounded to 3 digits)	–25.642 MPa
Intrinsic permeability	$1.15 \cdot 10^{-17} \text{ m}^2$
Porosity	0.231
Young's modulus	21.5 GPa

Listed are those parameters of the TRM model that were used for shotcrete and that are different from those used for concrete.

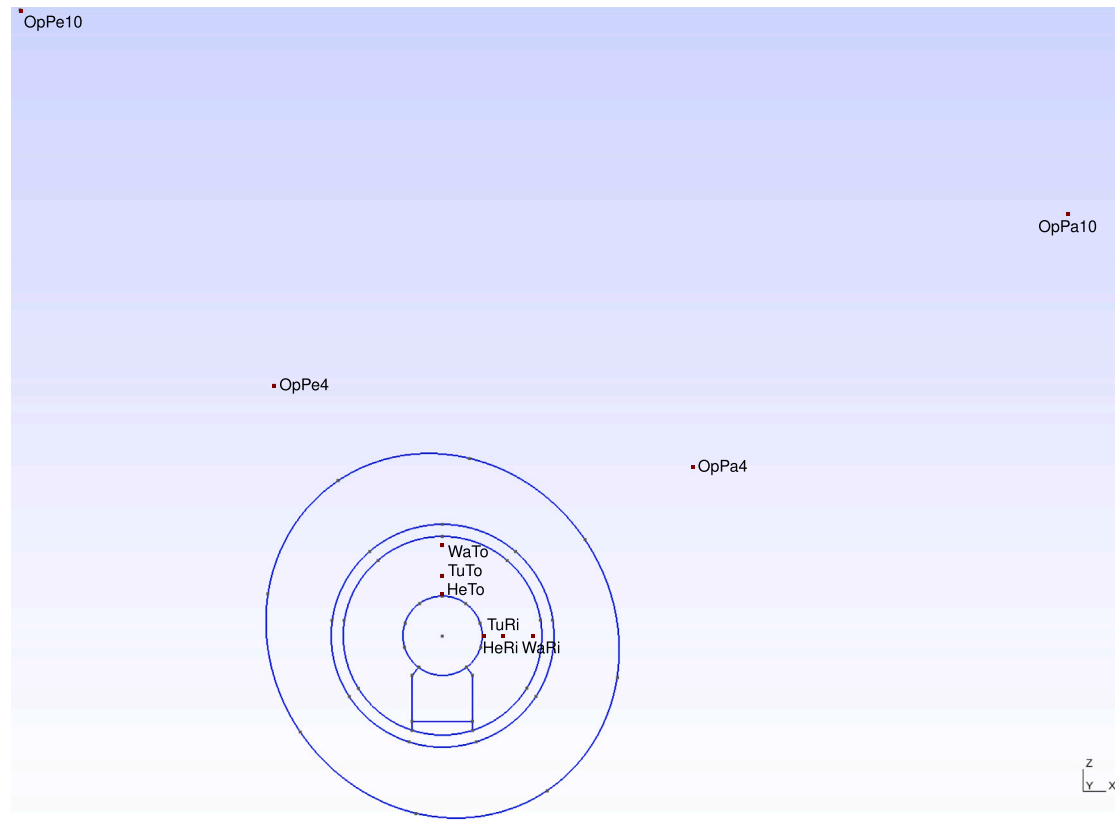


Fig. 4. Positions of those artificial measurement points, that were modified in each of the three spatial directions. All lie in the plane through the middle cross-section (i.e. xz plane) of the second heater. The geometry from outside to inside of EDZ, shotcrete, tunnel, pedestal base, pedestal and heater is included for reference.

excavation, the host rock will be damaged, causing changes in hydraulic and mechanical properties. However, it was out of the scope of this work to simulate the actual development of an EDZ by excavation modelling like e.g. Refs. 26, 27 did. Thus, we compared a model that included a simplified EDZ as additional material with its own material parameters to a model without an EDZ.

The dimensions of the used EDZ, its shape and boundary are simplified compared to what might be expected. Although there is limited knowledge about the EDZ around the FE experiment, Ref. 1 stated that it is larger where the tunnel is not supported by shotcrete compared to where there is shotcrete support. They reasoned this finding with the more ductile properties of the steel set support in the shotcrete-free tunnel section, which promoted the formation of an EDZ more compared to the shotcrete support. Ref. 4 provided reasoning for a greater extent of the EDZ normal to the bedding compared to parallel to it, which is due to the anisotropy of material properties and stress. We therefore used an ellipse-shaped cylinder as the outer boundary of the simplified EDZ, whose measures are given in Table 6. Furthermore, Fig. 2 gives an impression of its shape, whose inner boundary is the outer boundary of the combination of the shotcrete layer and the bentonite-filled tunnel.

The EDZ is simplified also in that sense, that we used a homogeneous material with linear elastic properties for it. Instead of explicit preferential flow paths or damage fields, material parameters different from those of intact Opalinus clay were used. Variations in parameters that started with the values of the Opalinus clay resulted in a parameterisation of the simplified EDZ. The parameter values differing from those of the Opalinus clay are also shown in Table 6. The value used for the intrinsic permeability in the simplified EDZ parallel to the bedding is 0.50 times the lower boundary documented by Ref. 2, while perpendicular to the bedding, the factor is 0.35.

Table 6

Model parameters used for the EDZ that differ from those used for OPA.

Parameter	Used value
Thickness $\parallel$ where Shotcrete	0.8 m
Thickness $\perp$ where Shotcrete	1.0 m
Thickness $\parallel$ where no Shotcrete	2.0 m
Thickness $\perp$ where no Shotcrete	2.5 m
Length at tunnel end (y-axis)	2.0 m
Dry bulk density (rounded to 3 digits)	2311.807 kg m ⁻³
Pressure IC	0.5 MPa
Intrinsic permeability $\parallel$	$5.00 \cdot 10^{-19}$ m ²
Intrinsic permeability $\perp$	$3.50 \cdot 10^{-19}$ m ²
Porosity	0.18
Young's modulus $\parallel$	3.6 GPa
Young's modulus $\perp$	1.8 GPa

Listed are those parameters of the TRM model that were used for the EDZ and that are different from those used for Opalinus clay. The symbols $\parallel$ and $\perp$ refer to the orientation with respect to the bedding plane.

2.6.3. Adding a ventilation phase

A third hypothesis was to enhance the development of the modelled liquid pressure by better capturing the initial liquid pressure field at the start of heating. During the preceding ventilation phase, drainage from the host rock into the tunnel would lower the liquid pressure and establish a gradient in the Opalinus clay.

To test this hypothesis, we compared model runs with a ventilation phase followed by the heating phase to those with just the heating phase. The parameters used for the ventilation phase before the heating are given in Table 7. The conditions modelled in the tunnel during the ventilation phase were constant.

2.6.4. Adding a non-homogeneous initial liquid pressure field

In situ, a strong alteration of the liquid pressure field of the host rock occurs not only during the ventilation phase, but also during the

Table 7
Model parameters used for the ventilation phase.

Parameter	Used value
Duration	2.5 a
Deactivated materials	Bent, Block, Steel, Concrete, but not Shotcrete, EDZ, OPA
BC Tunnel Temperature	16.5 °C
BC Tunnel Relative humidity	65 %
BC Tunnel Pressure (rounded to 3 digits)	−57.254 MPa
BC Tunnel Displacement	Dirichlet 0 m

Listed are those model parameters that were used for the ventilation phase.

excavation phase prior to that. As mentioned earlier, the simulation of the excavation phase was out of the scope of this work. But instead, a non-homogeneous initial liquid pressure field can act as a proxy for the hydraulic effect of the excavation phase at the beginning of the ventilation phase. And for that, we could estimate whether drainage and liquid pressure might be altered significantly during the heating phase.

So, we established a non-homogeneous initial liquid pressure field with the liquid pressure decreasing stepwise towards the tunnel. Since this is only a rough approximation, details are given in [Appendix A](#), but not discussed further. We then compared model runs with the non-homogeneous initial liquid pressure field to such with a spatially constant initial liquid pressure in the host rock using the value given by Ref. 16.

In total, for the studies of increased complexity, 151 runs have been performed with various combinations of effects.

3. Results and their discussion

The results of the geometry and increased complexity studies will be provided and discussed here.

3.1. Geometry studies

In the geometry studies, the influence on the modelled maximum temperatures at different materials and distances from the heaters was investigated. The effect of changes concerning the inner tunnel radius, the heater representation, the measurement point coordinates uncertainty, and the geometrical aspects of adding a shotcrete layer was examined. In these studies, time is not that relevant. Once the temperatures have reached a certain level, the curves are (almost) equidistant. For [Fig. 5](#), the maximum temperature differences were thus examined for (a), (b) and (d) for numerical reasons about 232.5 d after the start of the heating, while for (c) after 5 a.

3.1.1. Influence of the inner tunnel radius

As shown in [Fig. 5a](#), an increase in the inner tunnel radius by 7.5 cm (about 6 %) results in maximal 3.7 K warmer maximum temperatures at the locations provided by the specification. Although mainly the tunnel region is affected to such high changes, the difference in Opalinus clay is small, with at most 0.14 K colder maximum temperatures.

When the inner radius of the tunnel is increased, the tunnel itself is heated more strongly due to the higher insulation effect of the greater amount of unsaturated bentonite materials. Both types of bentonite have lower heat conductivity compared to that of the fully saturated host rock. The resulting greater heat accumulation in the tunnel causes the temperatures in the Opalinus clay to remain lower when the inner tunnel radius is increased. Due to the volume ratio between the host rock and the engineered structure, the influence on Opalinus clay is small compared to the influence on the tunnel.

3.1.2. Influence of the heater representation

In [Fig. 5b](#), the influence of the representation of the heaters as volumes compared to internal boundaries is shown for the same locations. Granular bentonite and Opalinus clay stay cooler with a reduction

of up to 16.6 K of the maximum temperatures on the heater surface in granular bentonite. But the bentonite blocks get warmer with a maximal increase of 12.6 K. Again, the difference in Opalinus clay is small, with a maximal decrease of 0.37 K. Thus, also with regard to heater representation, mainly the tunnel region is affected.

Representing the heaters explicitly with volumetric heat sources changes the temperature distribution between the two bentonite types compared to representing the heaters as holes combined with surface heat fluxes. The difference lies in the ability of the heater volumes to store and conduct heat. The additional heat storage results in lower temperatures in the granular bentonite because the latter has both low initial saturation and low heat conductivity. Consequently, we also find lower temperatures in the Opalinus clay. In addition to the insulating effect of the granular bentonite, the positions of the investigated points in the Opalinus clay are also worth mentioning here. They all lie in the upper half of the model domain and are thus hardly influenced by the bentonite blocks. However, the temperatures in the latter increase more when the heaters are represented as volumes compared to internal boundaries. This is caused by the bentonite blocks themselves, which have a higher initial saturation and thus conduct heat more efficiently than the granular bentonite. So, in the case of heater volumes, the heat transport is more effective in their direction.

If one is interested in the temperature distribution close to the heater, it may be worthwhile to better represent the internal structure of the heaters or, in the application case, the disposal cells. Some insights about the use of volumetric heaters with detailed internal structure can also be gained from the discussion of Step 1a by Guo and Briggs in Ref. 17.

3.1.3. Influence of the measurement point coordinates uncertainty

The influence of the uncertainty of the coordinates of the measurement points has first been investigated by comparing the two sets of coordinates described earlier. When using coordinates that as far as possible resemble those in reality in contrast to the more idealised ones by Ref. 16, the maximum temperatures drop. For the sensors directly at the heaters, the difference is at most 5.3 K, accompanying a location difference of up to 4.3 cm. The sensors 20 cm away from the heaters show the largest maximal difference with 8.3 K, associated with a difference in location of maximal 7.18 cm. At the sensors 10 cm away from the tunnel wall, the maximal difference is 7.6 K, together with a difference in location of up to 15.63 cm.

Most of the measurement point coordinates have been moved outwards when changing from the more idealised coordinates to those that resembled the ones in reality better. Larger radii and thus lower temperatures are the consequence. The farther away from the heaters and the smaller the difference in coordinates, the smaller the influence. However, since only measurement points with changed coordinates could be compared in this way, no information about the Opalinus clay could be gained from that comparison.

Artificial measurement point coordinates

Therefore, in a second step, the effect of the coordinate uncertainty has been investigated systematically by varying the coordinates of artificial measurement points in all three spatial directions. As indicated in [Fig. 5c](#), when testing the entire range from close to the heater to well within the Opalinus clay, strong gradients are visible, especially in the tunnel region. Once again, mainly this region is affected. A 10 cm change close to the heater leads e.g. to a maximal difference of 15.1 K in the maximum temperatures. However, the maximal difference in Opalinus clay – even with a change of 10 cm, already at a radius of 4 m, and perpendicular to the bedding – is still small with about 0.32 K.

As revealed by this systematic variation of artificial measurement points, the uncertainty of the sensor location is particularly relevant if strong gradients are present. For example, the effective isolation of the dry granular bentonite results in pronounced temperature gradients. Consequently, the coordinate uncertainty of the measurement points

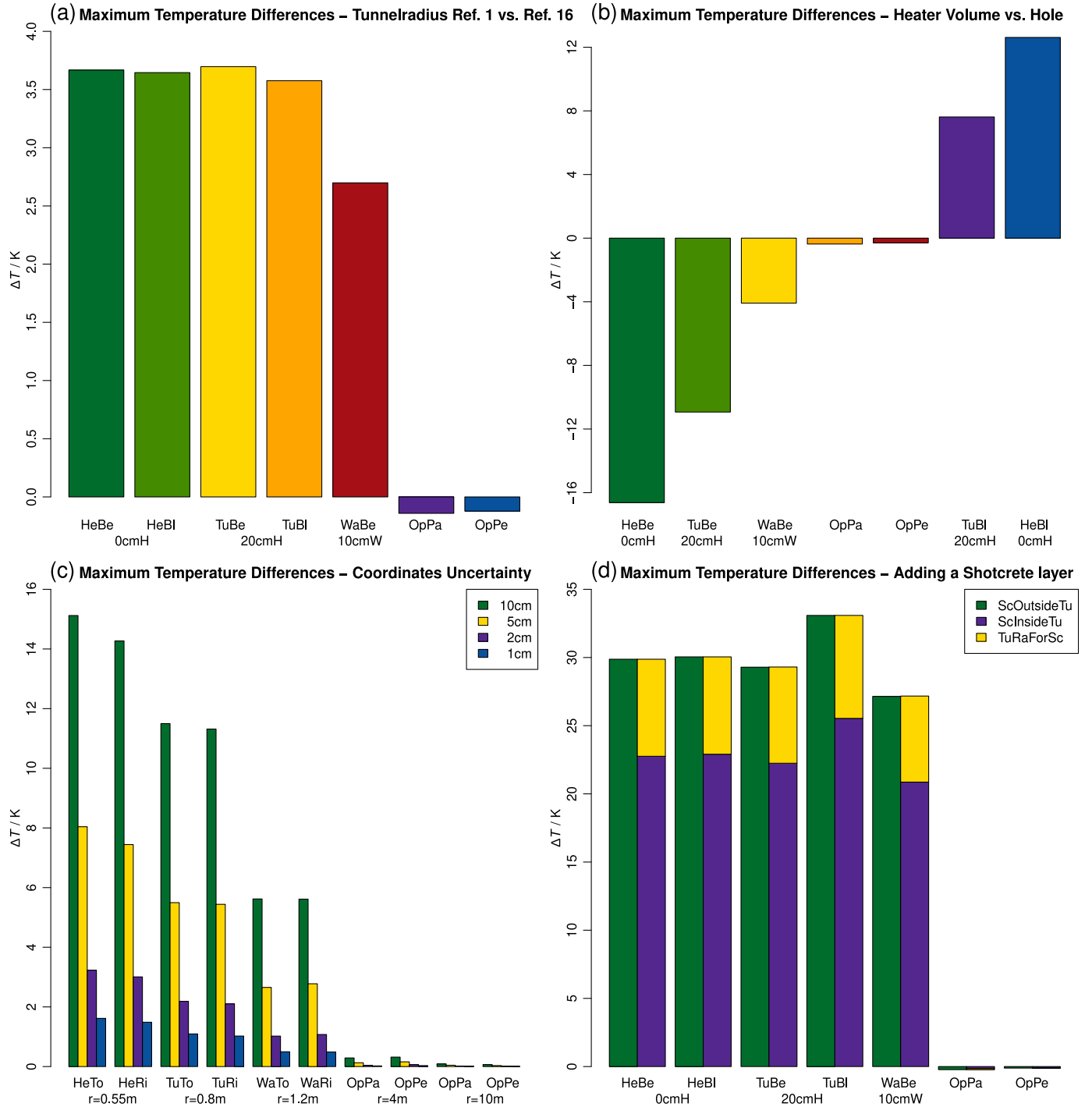


Fig. 5. Maximum temperature differences at different materials and distances from the heaters for (a) inner tunnel radius variation in line with Ref. 1 compared to Ref. 16, (b) representation of the heaters as volumes compared to holes with surface heat fluxes, (c) measurement point coordinates uncertainty, i.e. systematic variation of artificial measurement point coordinates in the three spatial directions by up to ± 10 cm compared to no variation, and (d) adding a shotcrete layer (with properties of concrete) outside vs. inside the tunnel and additionally adjusting the tunnel radius, compared to no shotcrete. The used categories for the measurement points in (a), (b) and (d) are those provided in Section 2.3 and Table 1 with their locations sketched in Fig. 3, while the artificial measurement points varied in (c) are those presented in Section 2.5.3 and Fig. 4. In (a) and (b), the colours have no meaning.

can be a contributor to a deviation from the measurement data when quantities with strong gradients are examined.

3.1.4. Influence of the geometrical aspects of adding a shotcrete layer

Finally, Fig. 5d shows the effect of the geometrical aspects of adding a shotcrete layer (with concrete properties) outside the tunnel vs. inside

the tunnel and additionally adjusting the tunnel radius, compared to the base case without shotcrete. As mentioned earlier, placing it outside the tunnel effectively reduces the volume of Opalinus clay, whereas placing it inside the tunnel replaces a portion of the bentonite's volume. Similarly, increasing the tunnel radius involves substituting some of the Opalinus clay's volume with bentonite.

In general, the addition of a shotcrete layer leads to an increase in the maximum temperatures at the locations provided by the specification. The maximal change is 33.1 K when adding a shotcrete layer (with concrete properties) outside the tunnel. But it is just 25.5 K when adding a shotcrete layer inside the tunnel, with a maximal additional change of 7.6 K for the adjustment of the tunnel radius. Both add up perfectly to adding a shotcrete layer outside the tunnel. Also in this scenario, the primary impact is on the tunnel, with only minor reductions observed in the Opalinus clay. The greatest decrease in the maximum temperatures there is 0.23 K parallel to the bedding, which occurs when a shotcrete layer is applied externally to the tunnel.

Adding a shotcrete layer leads to substantially higher temperatures in the tunnel for two reasons. The first is the use of concrete parameters which in sum imply higher thermal insulation capabilities compared to the other geotechnical materials present. As discussed earlier, this higher thermal insulation also leads to slightly lower temperatures in the Opalinus clay. However, again, this effect is small compared to the effect on the structure, because of the proportion of material there compared to the volume of Opalinus clay. The other reason is the increased amount of material in the structure itself, when shotcrete is added outside the tunnel or inside the tunnel with additionally adjusting the tunnel radius. We will see in the next section that almost only this part of the effect of adding a shotcrete layer remains, once it is equipped with more suitable parameters.

Overall, geometrical aspects such as the ones discussed above can become quite important, in particular in or close to the tunnel structure.

3.2. Increased complexity studies

In the studies examining increased complexity, the attention was primarily focused on the influence on the liquid pressure. The effects of the inclusion of shotcrete and a simplified EDZ as additional materials, a ventilation phase prior to heating, and a non-homogeneous initial liquid pressure field within the Opalinus clay were investigated, as well as various combinations of them. Using the findings of the geometry studies, appropriate dimensions and parameters for these new elements were incorporated as described in Section 2.6.

In general, the temperatures are not significantly influenced by the changes made in the increased complexity studies, even with the geometrical choices discussed above. Only adding a shotcrete layer (with the parameters in Table 5) leads to an increase in the maximum temperatures compared to the absence of shotcrete (data not shown, but see Appendix B, Fig. B.1 for some exemplary temperature results in OPA). But even when it is added outside of the tunnel, thus replacing Opalinus clay with shotcrete, the highest increase in the maximum temperatures compared to the absence of shotcrete is less than 8.5 K. It is located in the bentonite blocks 20 cm away from the heater. In the other measurement location categories, the maximum temperatures are up to 4.6 resp. 5.7 K higher at the heaters or 20 cm away from them, about 2.7 K higher 10 cm away from the tunnel wall in granular bentonite, and at most about 0.15 resp. 0.2 K higher in Opalinus clay parallel and perpendicular to the bedding. However, this is a substantial reduction in the increase in the maximum temperatures compared to using concrete parameters.

The variations carried out in the increased complexity studies (except for the addition of shotcrete) did not significantly influence the temperatures because the changes made do not relate to thermal properties of model parts. In case of shotcrete, in contrast to Section 3.1.4, the parameters in Table 5 were used for the investigations here. These alleviate the effect of a shotcrete layer on the temperature to almost only that part of it caused by the increased amount of material in the tunnel structure. This holds for both cases: when shotcrete is added outside of the tunnel, or inside with additionally adjusting the tunnel radius.

The relative humidity in the tunnel, in granular bentonite and bentonite blocks, is reduced by introducing the changes listed in the

increased complexity studies. In granular bentonite at the tunnel wall, the volumetric water content also decreases in those regions along the tunnel axis, where the heaters are located. However, it increases in the space between the heaters if a simplified EDZ is present and decreases if an EDZ is omitted (all not shown, but see Appendix C, Fig. C.1 for some exemplary relative humidity results in the tunnel in the granular bentonite). But since these effects are minor compared to the consequences for the liquid pressure in Opalinus clay, measurement data to compare to is of limited availability, and the uncertainty about the results in and close to the tunnel remains high, we focus here on the liquid pressure in Opalinus clay.

For simplicity, we present results for the liquid pressure at two measurement points in an exemplary way in Fig. 6. In Appendix D, however, Figs. D.1 to D.3 show results for all available liquid pressure measurement points in Opalinus clay. Both of the here selected measurement points sit at position 3 (out of 6) in their borehole. This means that they are approximately located in the region around the plane through the middle cross-section of the second heater. In detail, their y coordinates in the direction of the tunnel are about 33.0 m and 32.6 m respectively, while the heater spreads from 30.3 m to 34.9 m. Furthermore, both share almost the same radial distance from the y -axis through the tunnel centre of about 5.2 m, thus lie in Opalinus clay. The first point, corresponding to Fig. 6a, addresses the region parallel to the bedding, while the second point, corresponding to Fig. 6b, represents the region perpendicular to the bedding. This kind of information is given in the headers of the subfigures. In terms of their locations in the xz plane, a sketch is also included in Fig. 6. For even more detailed information, the reader is referred to Ref. 16 for the idealised measurement point coordinates. For the real ones, including location sketches for every single measurement point, refer also very well to Refs. 3, 5. As far as only points in Opalinus clay are concerned, both are nevertheless the same.

3.2.1. Influence of including shotcrete as additional material

We saw in Section 3.1.4 that adding a shotcrete layer has a larger effect when it is added by replacing some of the host rock with it compared to adding it by replacing some backfill material. To get an estimate of the maximal effect, we therefore present results here for adding it outside of the tunnel, thus replacing some Opalinus clay with shotcrete. Furthermore, as described in Section 2.6.1, here we used the parameters in Table 5 for shotcrete in contrast to those in Ref. 16 used in Section 3.1.4. With this configuration, the presence (—) or absence (—) of shotcrete alone does not have much effect on the liquid pressure. However, when added, a small damping is visible parallel to the bedding accompanied by a small increase in the perpendicular direction.

The addition of shotcrete had a relatively small influence on the liquid pressure because of its hydraulic properties. The saturation and porosity, for instance, have values between those of bentonite and Opalinus clay, while its intrinsic permeability is higher than that of all the other included materials. Consequently, the shotcrete layer establishes a hydraulically open transition zone. The small disbalance between the changes in the different directions originates from substituting the anisotropic Opalinus clay by the isotropic shotcrete. We then included shotcrete in all subsequent models.

3.2.2. Influence of additionally including a simplified EDZ

If to the base model not only shotcrete (—) but also a simplified EDZ (—) as described in Section 2.6.2 is added, the liquid pressure drops significantly. In particular, we find earlier and lower peaks and stronger subsequent decreases, especially in the vicinity of the tunnel and the EDZ. This is much less pronounced farther away from the structure. It is even slightly reversed in the case of some measurement points in parallel direction at radii larger than 9 m, where the liquid pressure has not yet peaked after 5 a of heating. However, in comparison to the change in the vicinity of the structure, these effects are small.

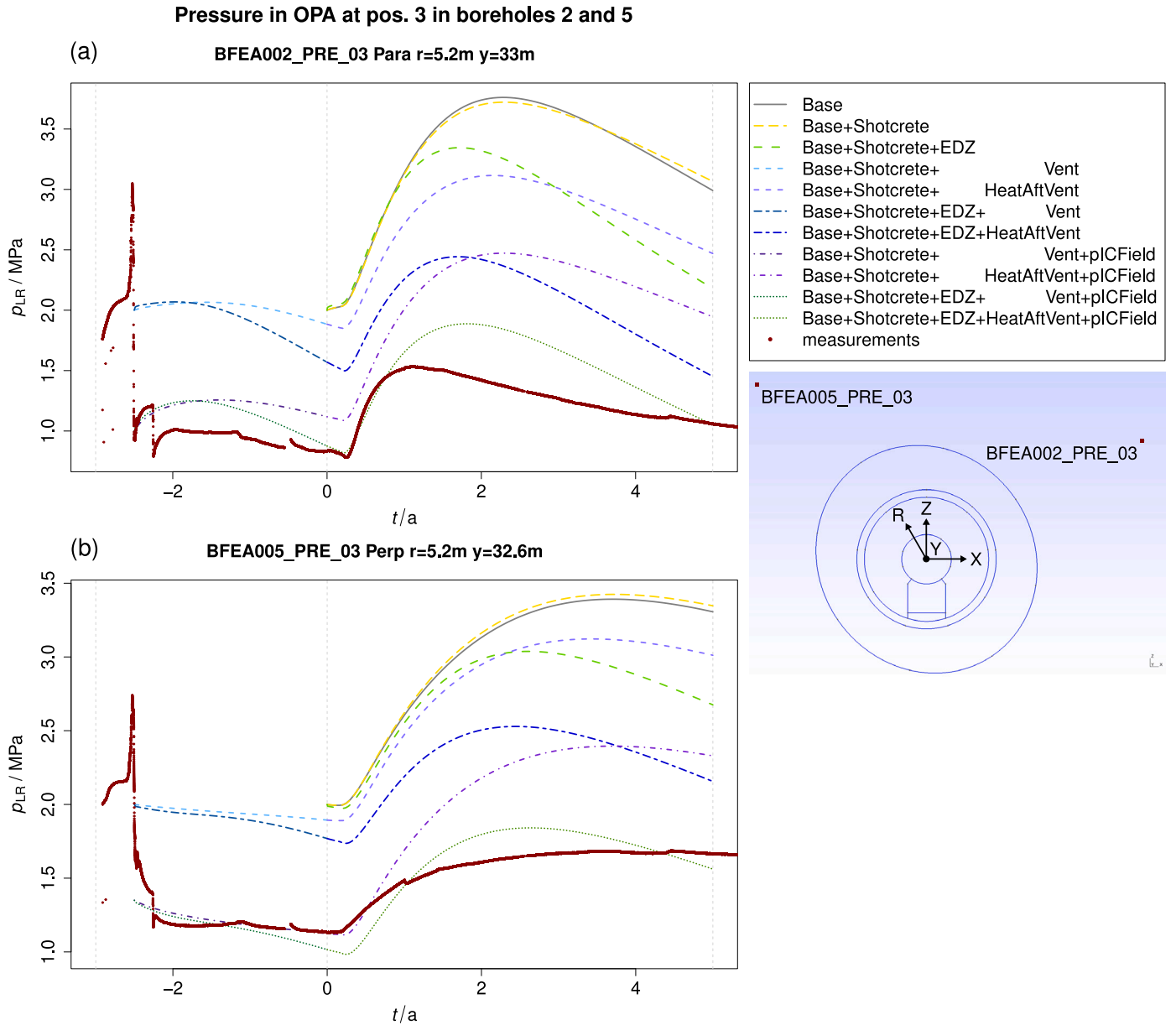


Fig. 6. Liquid pressure in OPA at position 3 in the boreholes (a) 2 parallel and (b) 5 perpendicular to the bedding at a radius of 5.2 m and in the direction of the tunnel approx. in the middle of the second heater. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. To the right, below the legend, the positions of the two selected measurement points are shown. As in Fig. 4, the geometry of the cross-section (i.e. xz plane) through the second heater is included for reference. The variables r and y in (a) and (b) refer to the radius and y -axis also shown there.

The reason for the significantly lower liquid pressure when a simplified EDZ is added additionally is due to the properties of the simplified EDZ. Especially its reduced initial liquid pressure and its increased porosity as well as intrinsic permeability, compared to those of the intact Opalinus clay, lead to this pattern. The influence of these changes propagates into the Opalinus clay with time, which is visible when comparing measurement points with increasing radii but otherwise similar positions. The slight increase in the liquid pressure in the parallel direction at some measurement points with large radii is due to the same reason. But as the enclosed volume increases with the radius, the effect is small compared to the changes at smaller distances.

3.2.3. Influence of including a ventilation phase additionally to a shotcrete layer

Adding a ventilation phase (---) before heating with the properties described in Section 2.6.3 in general only leads to small (<0.1 MPa) changes of the liquid pressure values at the beginning of heating compared to starting directly with heating (—). We see an increase parallel to and a decrease perpendicular to the bedding. Only the boreholes closest to the tunnel show a larger decrease for both directions (up to 1.05 MPa in the extreme, parallel to the bedding). During the heating following a ventilation phase, the liquid pressure evolves to

lower values and peaks, but also less steep post-peak decay (---) compared to without a ventilation phase (---).

The small effect of the ventilation phase at most of the measurement points is due to the low permeability of Opalinus clay and the relatively short duration of the ventilation phase. The reason for the small differences between the two directions at the onset of heating is probably their different hydraulic properties but equal initial values. Closer to the tunnel, the influence of the ventilation phase is larger due to higher driving gradients. As the liquid pressure could already decrease and equilibrate during the ventilation phase, the increase and development during heating is, in general, less pronounced compared to the case without a ventilation phase.

Compared to adding a simplified EDZ (---), the overall pattern of change is similar when adding a ventilation phase (---). Again, we find reduced peaks and a similar pattern of influence relatively seen when different points are compared. However, after a ventilation phase, the peaks are reduced in parallel direction more, in perpendicular direction less than if a simplified EDZ is used instead. Additionally, adding a ventilation phase results in a significantly smaller shift of the peaks towards earlier times.

The reason for the generally similar patterns between the addition of a simplified EDZ and a ventilation phase lies in the similarity between their effects. The ventilation phase allows pressure equilibration, thus also drawdown, and the simplified EDZ allows for accelerated drainage. The differences between both lie in the anticipation of relaxed conditions in the EDZ region, which still have to evolve when using a ventilation phase.

3.2.4. Influence of combining a shotcrete layer, a simplified EDZ and a ventilation phase

If both, simplified EDZ and ventilation phase (----), are present in addition to shotcrete (---), the liquid pressure in the vicinity of the structure is substantially reduced at the beginning of the heating. During heating, both changes add up to strongly reduced peaks, which are shifted to even earlier times compared to the individual contributions (--- and ---).

The reduction after 5 a of heating is stronger than the mere addition of both changes' reductions due to the still ongoing dissipation.

3.2.5. Influence of combining a shotcrete layer, a ventilation phase and a non-homogeneous initial liquid pressure field

The measurement data suggests an additional modification of the initial liquid pressure field in the Opalinus clay as a proxy for the consequences of excavation, which was not modelled here. The main effect of additionally using a non-homogeneous initial liquid pressure field (---) as described in Section 2.6.4 is an adjustment of the liquid pressure at the beginning of heating. However, the result is not a pure shift of the curves during heating. Instead, the deviation from the case with just shotcrete and ventilation phase (---) decreases over time. Moreover, the liquid pressure field equilibrates relatively quickly during the first year of the ventilation phase, transitioning to a steadily increasing radial field. As a consequence, the shapes of the curves during the ventilation phase differ when comparing cases with and without the non-homogeneous initial liquid pressure field.

With the non-homogeneous initial liquid pressure field used in the Opalinus clay, the liquid pressure is adjusted at the onset of heating. Although far from perfect, its use brought the model results substantially closer to the measurement data. The in general reduced values in the non-homogeneous initial liquid pressure field reduce the overall gradients between Opalinus clay and bentonite and thus moderate the overall hydraulic response.

An interesting comparison is simplified EDZ (----) vs. non-homogeneous initial liquid pressure field (---), both in combination with shotcrete and ventilation phase. During heating, a simplified EDZ reduces liquid pressure peaks and their timing relatively seen more than the non-homogeneous initial liquid pressure field. Only in the farthest

boreholes, the increase is larger at most measurement points when a simplified EDZ is added compared to a non-homogeneous initial liquid pressure field. But this larger increase is still small compared to the decreases closer to the tunnel.

During heating, the stronger impact of the simplified EDZ is due to its permanent presence, while the influence of initial values decays over time.

3.2.6. Influence of combining all four features together

The last of the models presented here combines all these changes: shotcrete, a simplified EDZ, a ventilation phase, as well as a non-homogeneous initial liquid pressure field (-----). In particular, two comparisons are worth mentioning here: Comparing this model to one without an EDZ (---) and comparing it to one with homogeneous initial liquid pressure (----). We found that when combined with the simplified EDZ, the effect of the non-homogeneous initial liquid pressure field is stronger. However, in combination with the non-homogeneous initial liquid pressure field, the effect of the simplified EDZ is weaker.

These observations have their reasons in the previously discussed influence of the simplified EDZ (Section 3.2.2) and the non-homogeneous initial liquid pressure field (Section 3.2.5), respectively. Still, the combination of all features does not lead to a mere superposition of their effects due to the non-linear thermal and hydraulic properties.

3.2.7. Comparing the most complex model to measurement data

The stepwise addition of model features significantly approached the modelled curves to the measurement data, albeit to varying degrees, depending on both the included features and the measurement points under consideration. However, even the most complex model (-----), i.e. the one with shotcrete, simplified EDZ, ventilation phase and non-homogeneous initial liquid pressure field, see Section 3.2.6, still shows deviations to the measurement data (•) at the measurement points studied. And while it compares best among the tested alternatives to the measurement data for most of the measurement point locations, there are also some locations, where other combinations give better results.

The reasons for this are numerous. There are, from one perspective, large spatial and temporal heterogeneities in the measurement data that were not modelled, even if the reason was known. From another perspective, there might be other processes or features that are missing in the model. We will discuss this in more detail in Section 4.

4. General discussion

The method described in Section 2.4 using the Index of Agreement helped significantly to assess the adequacy of the parameter values. It proved especially to be valuable, when comparing the results of different parameter combinations among each other at several measurement point locations and for several different variables simultaneously. Being able to objectively quantify and compare their quality of fit to the measurement data was a real benefit. Thus, we provided the used method in detail as a basis for systematic, objective and reproducible comparisons.

The influence of geometrical aspects like those discussed in Section 3.1 can be quite substantial, especially in and near the tunnel structure. The transfer from the geometrical representation to the results depends on the properties of the materials that occupy the respective parts of the domain. In other words, a geometrical change might be associated with a local change in material properties at an observation point, resulting in potentially substantial changes in the model results at that location. In our case, this larger influence was mainly concentrated in the region around the tunnel contour. The large volume of Opalinus clay outside this region was only slightly influenced. Furthermore, the geometry used as a whole was simplified compared to the real one, whose modelling was beyond the scope of this work.

Other features or processes that we did not consider in this work may have similar effects, but we do not want to speculate about their quantitative extent without having tested them. From a numerical point of view, even with the 2.91 M elements in the mesh, the resolution might still not be fully converged yet, especially around material interfaces with strong property contrasts.

The combination of several features as described in Section 3.2, that had only limited individual influence, yielded a quite acceptable comparison to the measurement data. To be precise, combining a shotcrete layer with a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field to account for excavation-induced effects led to a substantial liquid pressure reduction in the Opalinus clay during heating compared to models without these features. In Refs. 17, 28, the results of Section 3.2.6 are also compared to the results of other teams modelling the same experiment, giving a broad overview of the task in which the work at hand was integrated.

Despite the improvements due to the inclusion of the mentioned features in the model, differences to the measurement data remain. The potential reasons for this are the significant simplifications that persist. In addition to geometrical simplifications and the use of mostly homogeneous material property fields, these include especially the simplified representation of the EDZ (see Section 2.6.2) as well as of bentonite. The latter was represented as an elastic material, not taking into account compaction and swelling behaviour. This simplification affects both the hydraulic and mechanical response of the materials, and consequently also the thermal response to some degree.

However, there are spatial and temporal heterogeneities in the measurement data that have not been modelled here. These are likely due to excavation and its consequences such as shear bands, plasticity, cracks, fissures, etc. (see e.g. Refs. 29, 30). These features alter the stiffness and permeability around the excavation in a complex way. By simulating excavation itself, using for example elastoplasticity, damage and associated permeability changes, more realistic liquid pressure and property fields could be achieved.²⁹ The weather in the tunnel during the ventilation phase was also not considered. The model boundary conditions were chosen to be relatively simple and did not account for the structures and activities in the vicinity of the FE experiment. Features such as those mentioned are known to influence the measurements, contributing to the uncertainty described in Section 2.3.

5. Conclusions

The overall aim of this work was to model the FE experiment numerically and to reproduce its measurement data as well as possible. For this purpose, a stepwise procedure was adopted where in each step several hypotheses were examined to systematically assess their effects on the model results. The approach utilised the theory of non-isothermal Richards flow with mechanics in the form implemented in OpenGeoSys 6. Large detailed 3D meshes of up to 2.38 Mdof were used for extensive model variations.

We explored ways to incorporate large amounts of measurement data into the model evaluation process. Both the size of the 3D models and the amount of comparisons posed significant challenges. A procedure based on the Index of Agreement was established, which proved to be a valuable method for systematic, objective and reproducible comparisons. It thus enabled a quality of fit assessment of different model versions against a comprehensive set of measurement data.

The investigated hypotheses encompass both geometrical considerations and the complexity of the model itself. The geometrical considerations explored include the inner radius of the tunnel, the representation of the heaters, the variability due to uncertainty in the measurement point coordinates, and geometrical aspects related to the incorporation of a shotcrete layer. Model complexity has been increased by introducing shotcrete and a simplified excavation damaged zone (EDZ) as additional materials, incorporating a ventilation phase, and considering

a non-homogeneous initial liquid pressure field due to excavation effects.

The significant influence of geometrical aspects on the results in the tunnel structure became evident. Furthermore, the strong impact of the initial liquid pressure field and the intrinsic permeabilities in Opalinus clay and EDZ points to the importance of excavation modelling even in simulations of heater experiments. Integrating the above listed features has already greatly improved the alignment between model results and a substantial amount of measurement data.

Research efforts such as those discussed in this paper support work towards validating models. But additionally they enhance our understanding of physical processes and clarify uncertainties associated with the analysis of in situ systems. Such investigations can guide both modelling endeavours and discussions with experimentalists, e.g. concerning aspects of sensor positioning.

To model the immediate vicinity of excavations that include supporting structures and are backfilled with complex materials, while considering heating and two-phase flow processes, requires consideration of a broad spectrum of processes and material interactions. Particularly, we conclude that the integration of a non-homogeneous elastoplastic EDZ along with the swelling behaviour of bentonite in the model may further improve its ability to reproduce the measurement data.

CRedit authorship contribution statement

Sonja Kaiser: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Wenqing Wang:** Writing – review & editing, Software. **Jörg Buchwald:** Writing – review & editing, Software. **Dmitri Naumov:** Writing – review & editing, Software. **Aqeel Afzal Chaudhry:** Writing – review & editing, Software. **Thomas Nagel:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Details of the non-homogeneous initial liquid pressure field

The non-homogeneous initial liquid pressure field served as a simplified representation of initial liquid pressure alterations due to excavation. It consisted in its final form of nested ellipse shaped cylinders of constant liquid pressure, starting at the outer boundary of the EDZ and ending with the bulk domain of Opalinus clay. The initial liquid pressure value for the region where the EDZ would have been is still the one for EDZ, 0.5 MPa, but otherwise no properties of an EDZ are used. Built manually without proper excavation modelling, this initial liquid pressure field is still just a rough approximation. It aimed to approach the liquid pressure data at the start of the heating phase while being applied at the beginning of the ventilation phase. Its parameters are given in [Table A.1](#). Including the finally used one, 10 different non-homogeneous initial liquid pressure fields have been tested, likewise using an initial temperature field including a geothermal gradient.

Table A.1

Liquid pressure values and outer boundaries of their application cylinders in the finally used initial liquid pressure field for Opalinus clay.

Used Pressure /MPa	Semi Axis $\parallel$ /m	Semi Axis $\perp$ /m	Y Axis /m
1.000	5.45	4.10	59.40
1.500	9.40	5.05	61.85
1.350	9.40	5.30	61.85
1.300	9.40	5.55	61.85
1.000	9.40	7.65	61.85
1.375	9.40	8.35	61.85
1.250	9.40	9.05	61.85
1.000	9.40	10.00	61.85
1.900	9.40	11.30	61.85
1.850	10.00	12.70	62.85
1.900	10.90	12.70	63.75
1.950	16.50	12.70	69.35
1.800	16.50	14.05	69.35
1.750	16.50	18.00	69.35
2.000	max	max	max

Listed are the liquid pressure values as well as outer boundaries of their application cylinders in the finally used initial liquid pressure field for Opalinus clay. The symbols $\parallel$ and $\perp$ refer to the orientation with respect to the bedding plane. Y is parallel to the tunnel length.

Appendix B. Results for some of the available temperature measurement points

See [Fig. B.1](#).

Appendix C. Results for some of the available relative humidity measurement points

See [Fig. C.1](#).

Appendix D. Results for all available liquid pressure measurement points

See [Figs. D.1–D.3](#).

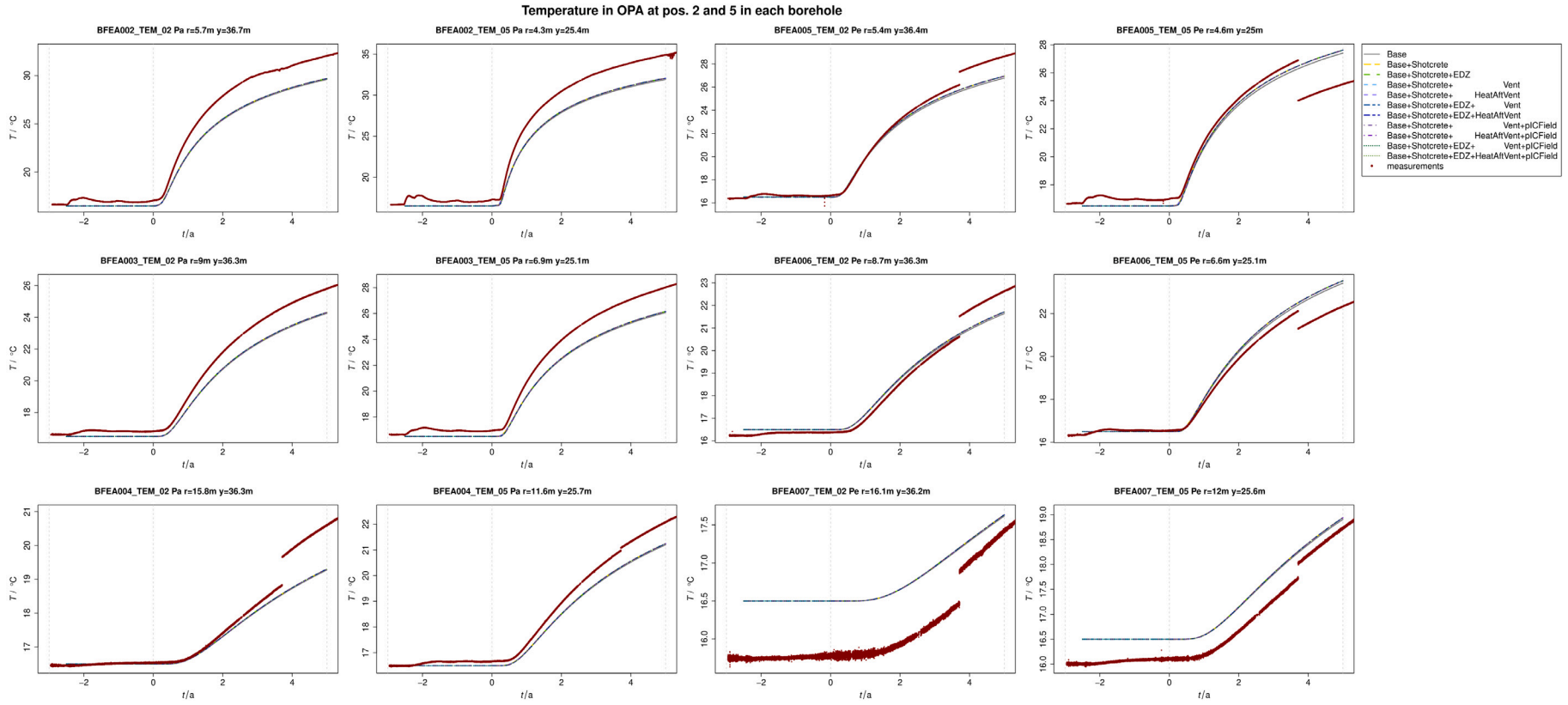


Fig. B.1. Temperature in OPA at positions 2 and 5 in each borehole. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. As in Fig. 6, the variables r and y refer to the radius and y -axis as shown there.

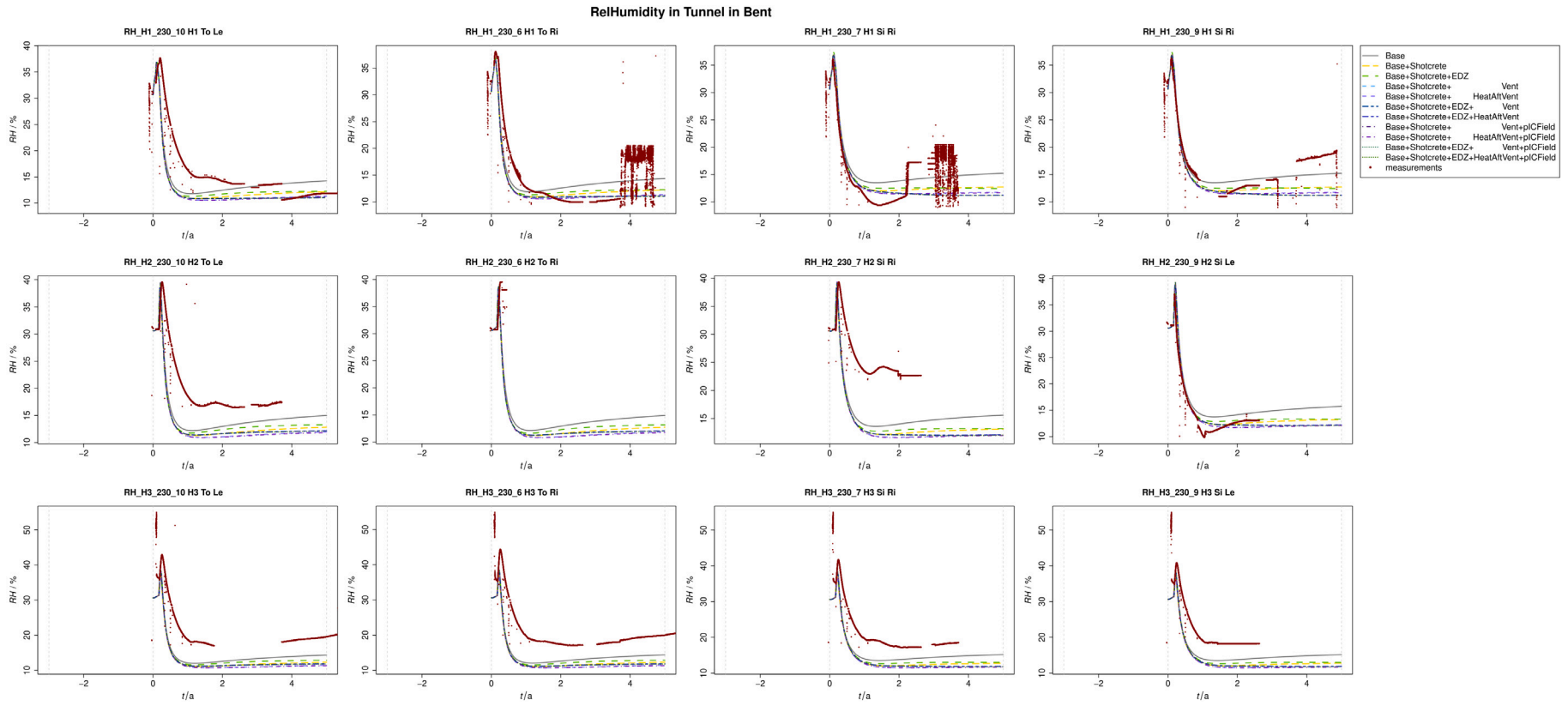


Fig. C.1. Relative humidity in the tunnel in the granular bentonite. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. As in Fig. 6, the variables r and y refer to the radius and y -axis as shown there.

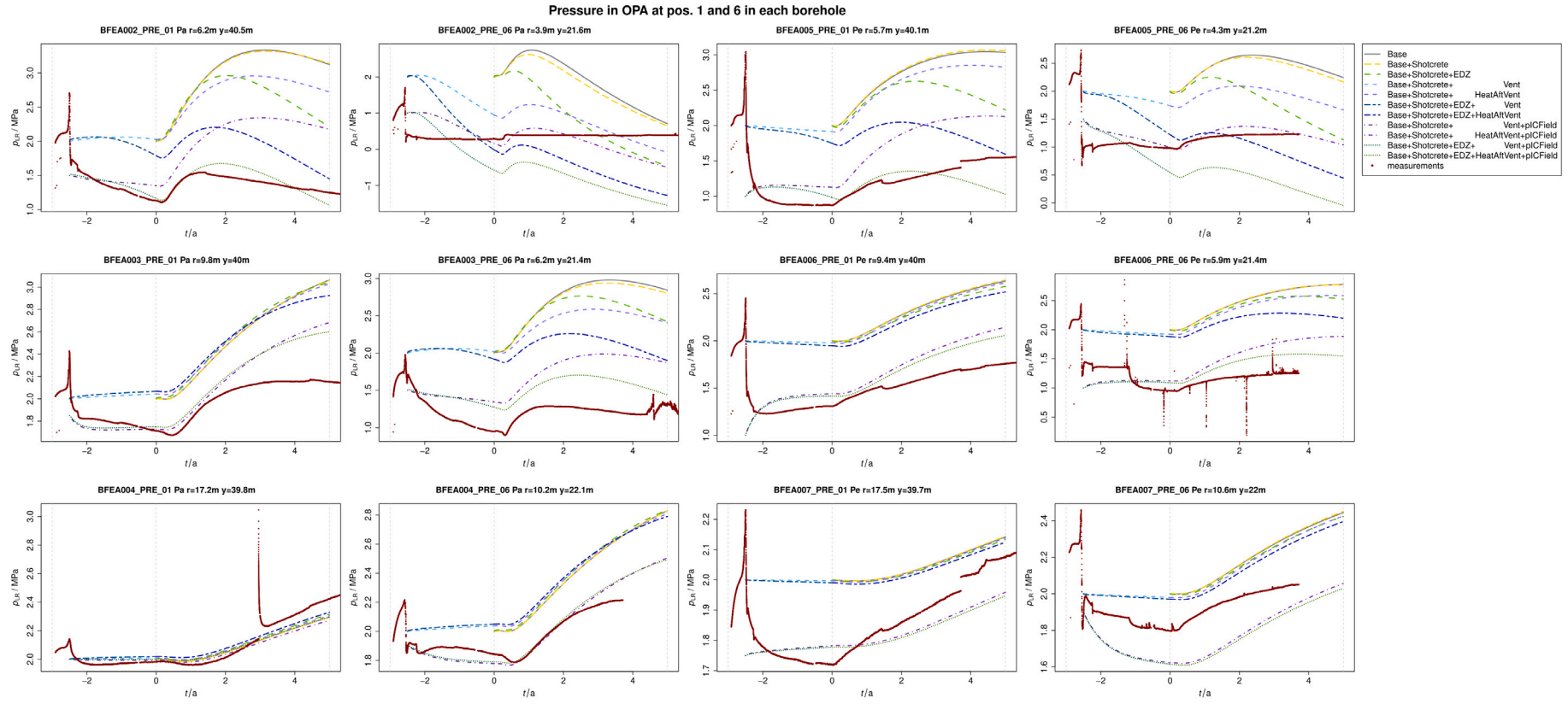


Fig. D.1. Liquid pressure in OPA at positions 1 and 6 in each borehole. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. As in Fig. 6, the variables r and y refer to the radius and y -axis as shown there.

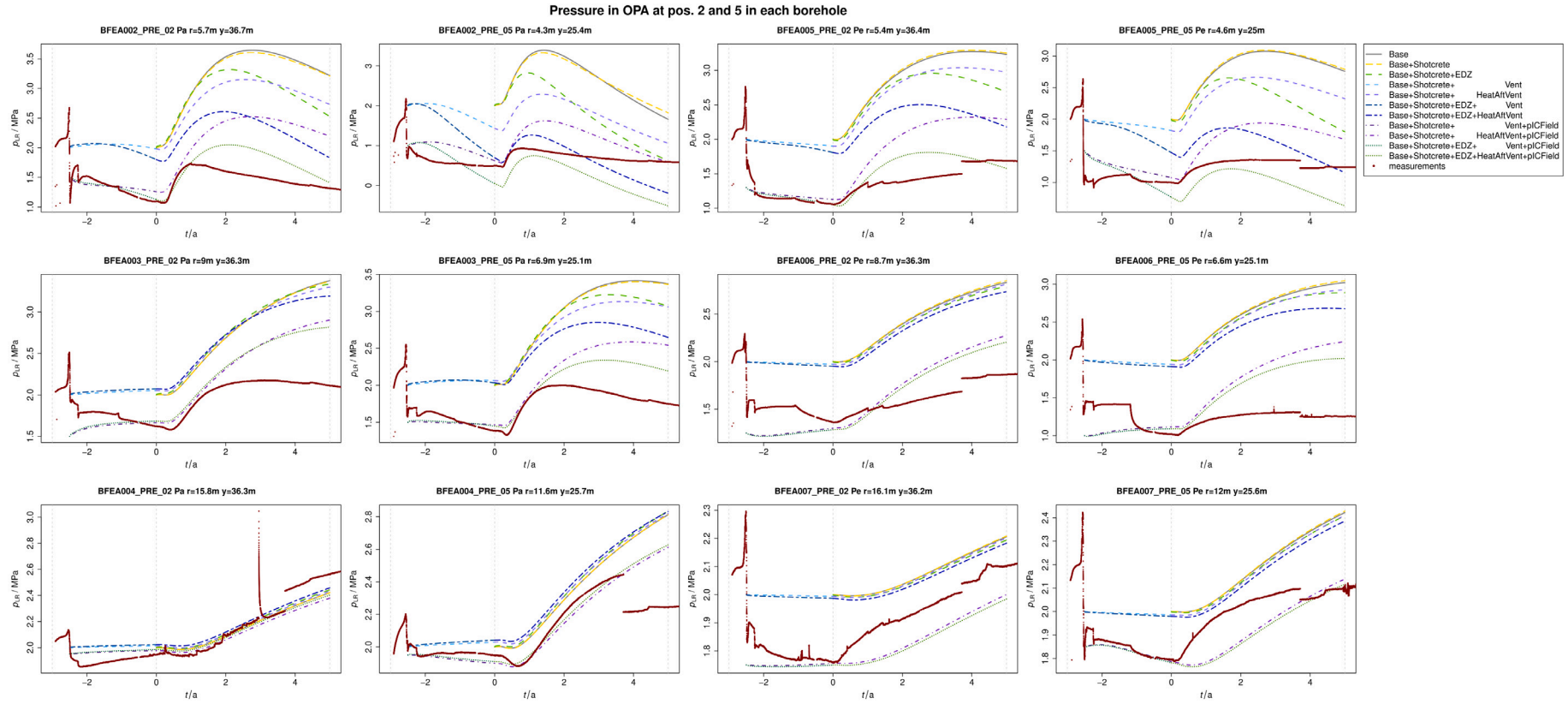


Fig. D.2. Liquid pressure in OPA at positions 2 and 5 in each borehole. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. As in Fig. 6, the variables r and y refer to the radius and y -axis as shown there.

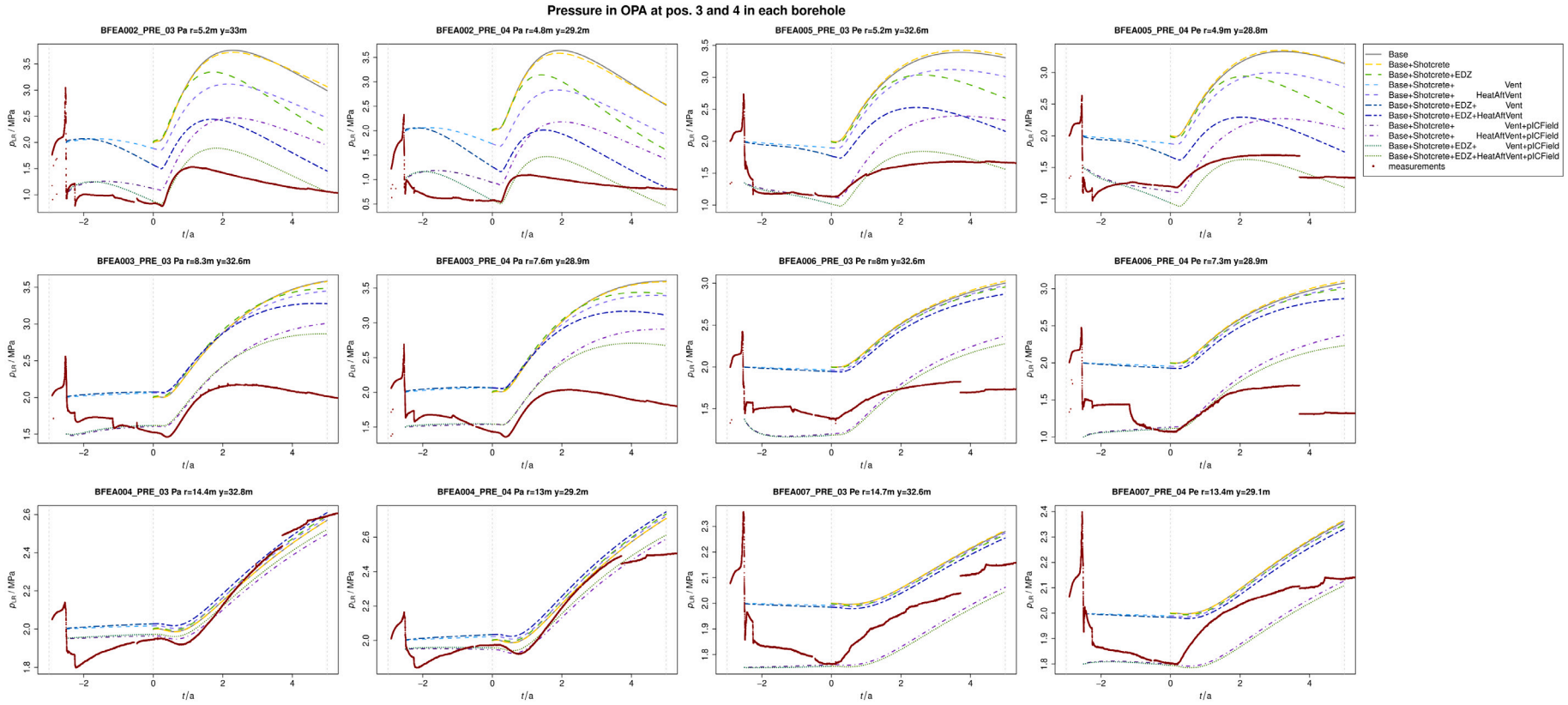


Fig. D.3. Liquid pressure in OPA at positions 3 and 4 in each borehole. The base case (with OPA in place of shotcrete and EDZ) is shown in comparison to runs with increasing complexity by adding shotcrete, a simplified EDZ, a ventilation phase, and a non-homogeneous initial liquid pressure field for OPA, as well as several combinations. Heating starts at $t=0$ a and red points mark the measurement data. As in Fig. 6, the variables r and y refer to the radius and y -axis as shown there.

Code availability

OGS-6 is available at <https://www.opengeosys.org/>.

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